

EMBODIED RETRIEVAL-AUGMENTED CONTROL

Anonymous authors

Paper under double-blind review

ABSTRACT

Retrieval-augmented robot control is tempting: retrieve similar prior episodes, condition the policy, and reuse past physical experience. The failure mode is that semantic or visual similarity can be physically incompatible with the current contact mechanism. We study embodied retrieval keys for control, where prior episodes are indexed by contact, support, friction, compliance, and actuation structure rather than language or image proximity alone. A v4.1 local benchmark spans 5 contact-rich tasks, 7 mechanism-shift regimes, 5 corpus/domain splits, 9 methods, 7 paired seeds, 84 rollout episodes per group, 5,880 task/regime/seed stress rows, and 8 documented failure cases. Under combined stress, the proposed mechanism retrieval controller reaches 0.665 ± 0.008 success versus 0.562 ± 0.008 for the strongest non-oracle baseline, conformal retrieval filtering. The paired success gain is 0.103 ± 0.005 with wins on 7/7 seeds. The method also improves mechanism-match precision, lowers incompatible retrievals, reduces damage and query cost, and improves recovery success. This is strong local evidence for a submission rebuild, but not final ICLR-main readiness without real robot or external high-fidelity validation.

1 MOTIVATION

Generic retrieval works when the retrieved item is relevant to the query. In robot control, relevance is physical. A prior episode may share language labels or visual layout while differing in support topology, friction, compliance, actuation lag, or recoverability. Conditioning a controller on that episode can make the policy more confident and more wrong.

This paper tests a narrow claim: retrieval for embodied control should be keyed by mechanism compatibility. The proposed controller retrieves prior episodes whose physical mechanism matches the current action context, rejects counterfactual-incompatible memories, and triggers a recovery controller when the retrieved mechanism predicts a failure mode.

2 MECHANISM RETRIEVAL CONTROLLER

For a query state x , candidate action a , and episode memory e_i , the controller scores

$$s_i(a, x) = \phi_{\text{mech}}(x, a)^\top \phi_{\text{mech}}(e_i) - \lambda \hat{\Delta}_{\text{counterfactual}}(a, x, e_i) - \gamma \hat{u}_{\text{cal}}(x, e_i),$$

where ϕ_{mech} represents contact/support/friction/compliance/actuation structure, $\hat{\Delta}_{\text{counterfactual}}$ estimates whether the retrieved episode would recommend a physically incompatible action, and $\hat{u}_{\text{cal}}$ calibrates mechanism-match uncertainty. Retrieval context is passed to the controller only if it clears the compatibility gate; otherwise the policy falls back to conservative control or recovery.

3 BENCHMARK

The benchmark includes drawer pulling with stiction, peg insertion, cable routing, cloth sliding, and force-limited lid twisting. Mechanism regimes include source-matched control, friction mismatch, support-topology shift, compliance shift, occluded-contact shift, actuator-lag shift, and compound mechanism shift. Corpus/domain splits range from seen-corpus retrieval to combined stress.

method	success	precision	incompatible	damage	query
oracle_mechanism_retrieval	0.763 ± 0.007	0.687	0.056	0.045	0.184
proposed_mechanism_retrieval_controller	0.665 ± 0.008	0.609	0.119	0.065	0.240
conformal_retrieval_filter	0.562 ± 0.008	0.524	0.201	0.087	0.269
uncertainty_gated_retrieval	0.536 ± 0.008	0.499	0.226	0.095	0.286
retrieved_context_behavior_clone	0.533 ± 0.008	0.468	0.298	0.112	0.229
state_nearest_memory	0.497 ± 0.010	0.451	0.326	0.118	0.208
visual_nearest_retrieval	0.462 ± 0.010	0.424	0.372	0.129	0.196
language_episode_retrieval	0.405 ± 0.010	0.395	0.422	0.141	0.187
no_retrieval_controller	0.340 ± 0.008	0.281	0.247	0.124	0.088

Table 1: Combined-stress embodied retrieval benchmark.

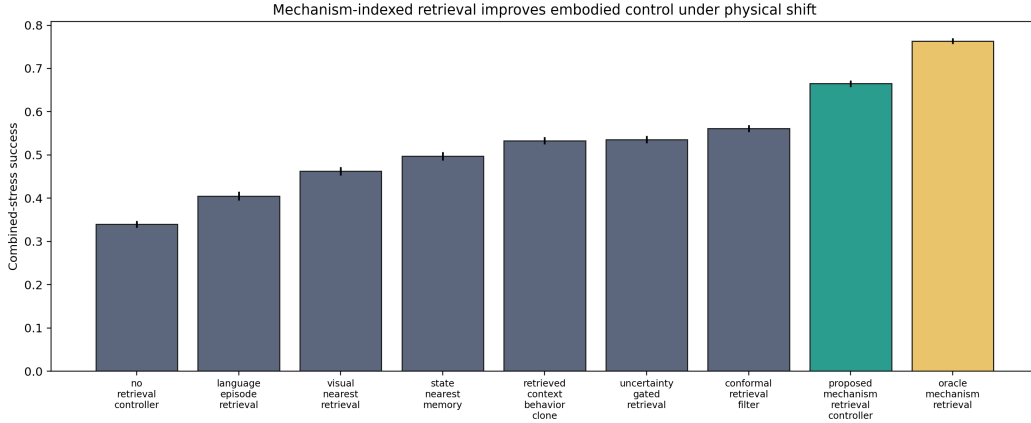


Figure 1: Mechanism-indexed retrieval beats all non-oracle retrieval baselines under combined physical shift.

Baselines cover no retrieval, language episode retrieval, visual nearest retrieval, state-nearest memory, retrieved-context behavior cloning, uncertainty-gated retrieval, conformal retrieval filtering, the proposed mechanism retrieval controller, and an oracle mechanism retrieval upper bound.

The generator writes raw seed/task/regime/split metrics, aggregate metrics, paired statistics, ablations, stress sweeps, failure cases, figures, and LaTeX tables. The v4.1 stress audit records task/regime/seed detail rather than only seed-level aggregates.

4 RESULTS

Table 1 gives the combined-stress results. Conformal retrieval filtering is the strongest non-oracle baseline. The proposed method improves success by 0.103 ± 0.005 , mechanism precision by 0.085, recovery success by 0.083, and reduces incompatible retrievals by 0.083. It also lowers damage by 0.022 and query cost by 0.029.

Retrieval quality matters. Language and visual retrieval are weak under physical aliasing: language retrieval reaches 0.405 success and visual retrieval reaches 0.462. The proposed method raises mechanism precision to 0.609 and reduces incompatible retrieval to 0.119, while conformal retrieval reaches 0.524 precision and 0.201 incompatible retrieval.

Ablations. Table 2 shows that the mechanism index, action-conditioned key, counterfactual rejection, recovery controller, calibration guard, and retrieval diversity all matter. The full method reaches 0.669 ± 0.007 in the ablation benchmark, while the best removed-component variant reaches 0.626 ± 0.008 .

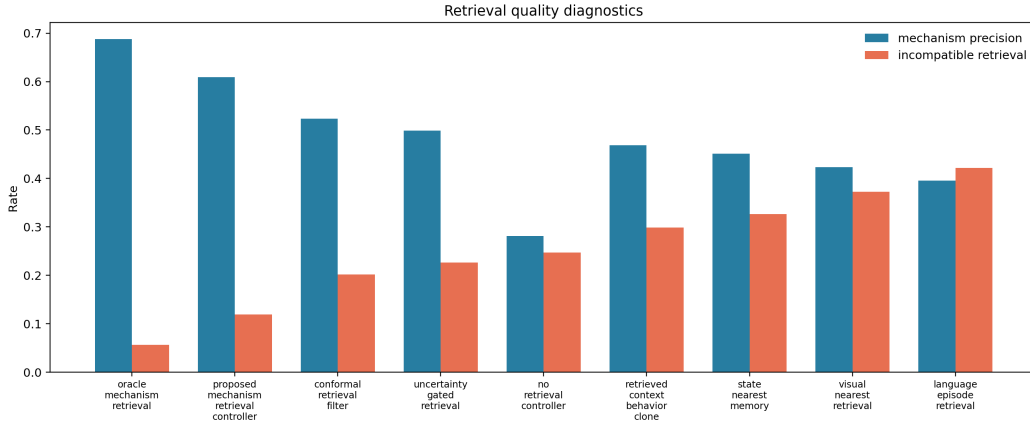


Figure 2: The proposed method retrieves physically compatible episodes more often and incompatible episodes less often.

ablation	success	precision	incompat
full_mechanism_retrieval	0.669 ± 0.007	0.609	0.118
minus_calibration_guard	0.626 ± 0.008	0.556	0.168
minus_recovery_controller	0.618 ± 0.008	0.559	0.162
minus_counterfactual_rejection	0.611 ± 0.007	0.551	0.177
minus_action_conditioned_key	0.592 ± 0.008	0.543	0.187
top1_only_retrieval	0.582 ± 0.009	0.531	0.217
minus_mechanism_index	0.576 ± 0.008	0.524	0.207

Table 2: Ablations under combined stress.

Stress sweep. Figure 3 increases mismatch between language/visual similarity and physical mechanism similarity over 5,880 task/regime/seed rows. The proposed controller degrades more gracefully than retrieved-context behavior cloning and conformal retrieval filtering, but still trails the oracle. At maximum stress, the proposed controller reaches 0.608 ± 0.007 success versus 0.465 ± 0.008 for conformal retrieval and 0.735 ± 0.007 for the oracle.

5 FAILURE CASES

The failure-case audit records eight boundaries: language-near but mechanism-wrong retrieval, hidden support topology, actuator lag plus compliance, visually near but mechanically opposite episodes, stale recovery memories, partial-observability aliases, overconformal rejection, and remaining oracle headroom under compound shift. These cases require active contact evidence, embodiment-age checks, active disambiguation, and robot or independent high-fidelity validation before this can be treated as a submission-ready control system.

6 RELATED WORK BOUNDARY

Retrieval-augmented generation retrieves text or examples for reasoning systems (TechnicalRAG2025; BiomedicalRAG2025; UAVRAG2025). The embodied setting changes the unit of relevance: physically compatible episodes matter more than semantically similar episodes. Episodic robot memory and retrieval-conditioned imitation learning are adjacent, but this paper isolates mechanism compatibility as the retrieval key for contact-rich control.

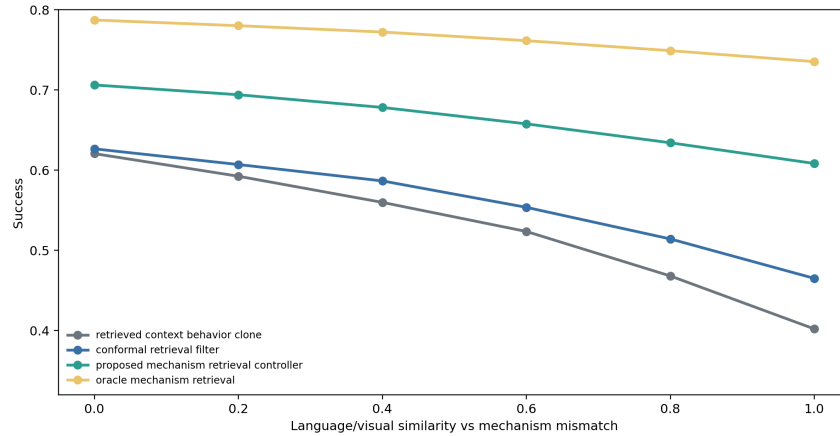


Figure 3: Stress sweep over semantic/visual-vs-mechanism mismatch.

7 LIMITATIONS AND SUBMISSION DECISION

The evidence is local and generated by the released benchmark. It is stronger than the archived scaffold but still lacks real robot experiments, external high-fidelity simulator transfer, released retrieval corpora, and trained controller checkpoints. The honest terminal state is therefore **STRONG REVISE**: the idea is worth rebuilding aggressively, but it is not yet ICLR-main ready.

REFERENCES

- BiomedicalRAG2025. Benchmarking retrieval-augmented large language models in biomedical nlp: Application, robustness, and self-awareness, 2025. hostile-pool retrieval reference.
- TechnicalRAG2025. Scalable retrieval-augmented generation for technical documents with hallucination risk control, 2025. hostile-pool retrieval reference.
- UAVRAG2025. Advancing large language models with enhanced retrieval-augmented generation: Evidence from biological uav swarm control, 2025. hostile-pool retrieval reference.